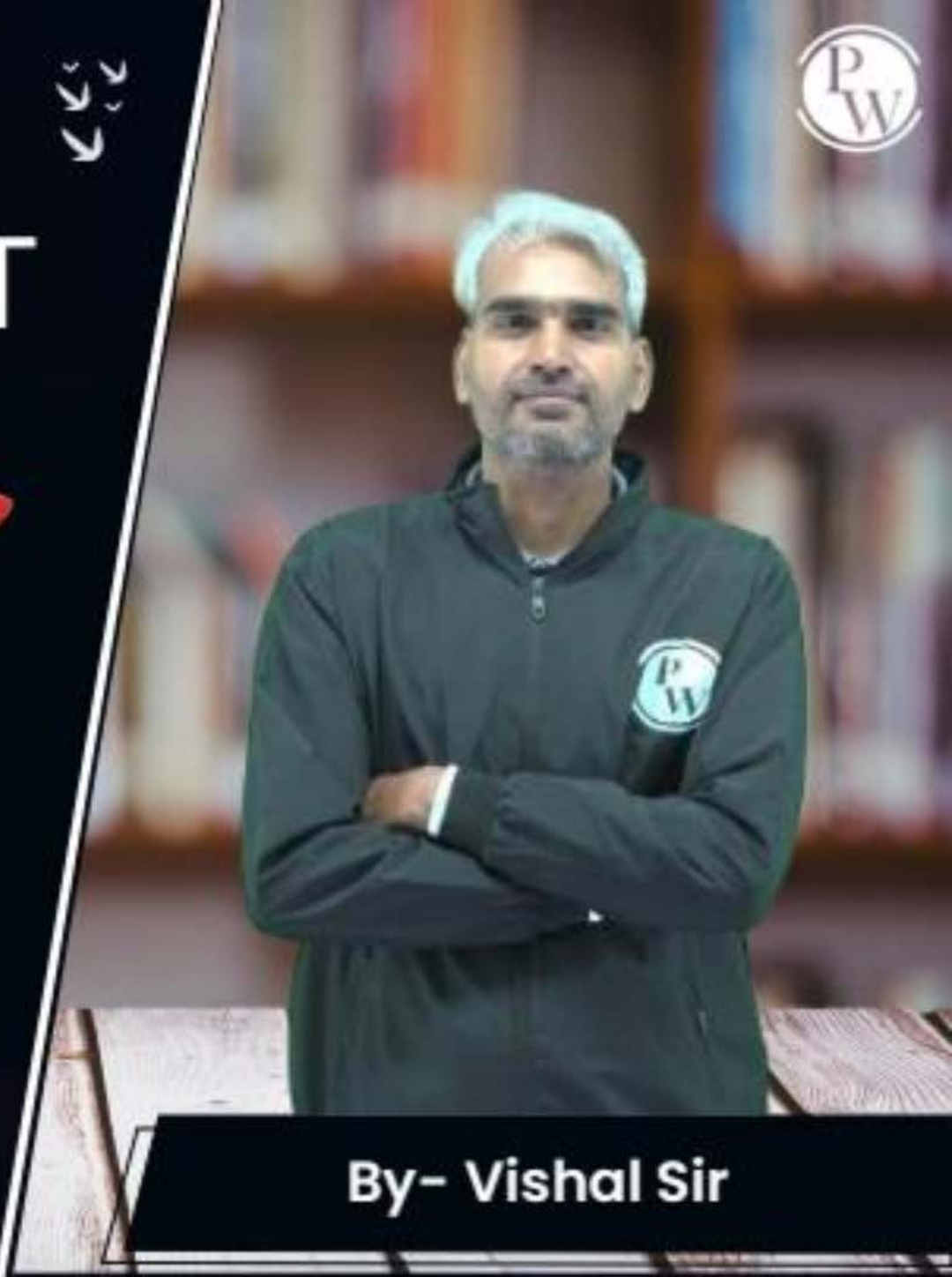
Computer Science & IT



Database Management System

Query Languages

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture



Topic Properties of multivalued dependency

Topic Fourth normal form (4NF)

Slide

Topics to be Covered



- Topic: Query Languages
- Topic: Relational algebra
- Topic: Basic relational algebra operations

Slide



Topic : Query languages

Query languages

Procedural query language

We define the procedure to retrieve the required information from the database

Eg: Relational Algebra

Non-procedural query language

We use the syntax provided by query language to retrieve the required information from the database

Eg:
- Structured Query language (SQL)
- Tuple Relational Calculus (TRC)
- Domain Relational Calculus (DRC)

Note :- Relational algebra query condition Evaluates tuple by tuple, taken only one tuple at a time.

∴ If we want to compare two or more tuples of the same table or different tables, then we need combine those tuples into a single tuple using "Join operation".

⇓

Eg Cross Join (Cross Product), Natural Join or any other type of Join opn

Note:- Relational algebra query will always produce distinct tuples.

{i.e., duplicate tuples will never be present in the output of Relational algebra query}



Topic : Relational Algebra

- Relational Algebra is a procedural query language used to query the relational database tables to access data.

- Relational Algebra operation can be classified into two types:
  1) Basic Relational Algebra Operations
  2) Derived Relational Algebra operations



Topic : Basic Relational Algebra operators

1. Projection(π)
2. Selection (σ)
3. Cross Product (×)

already discussed during Normalization

4. Union (U)
5. Set Difference (−)

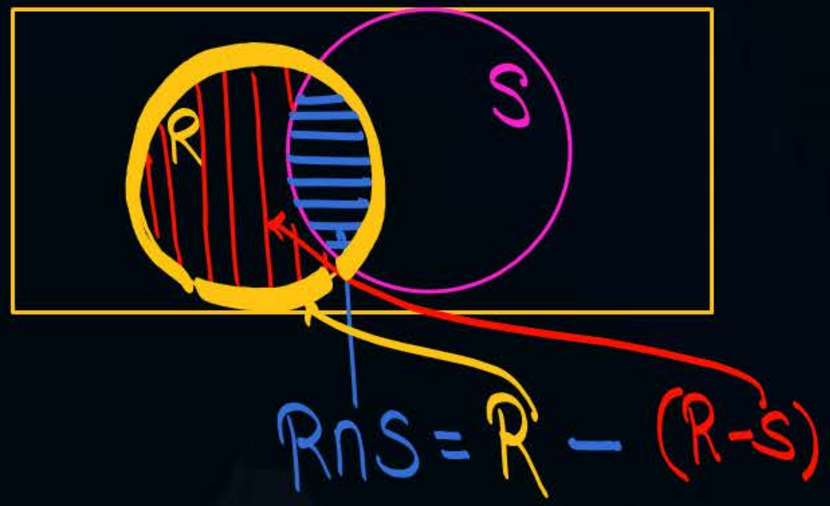
Set operations

6. Rename (ρ)



Topic : Derived Relational Algebra operators

1. Intersection (∩)
2. Join Operations ("⋈")
3. Division Operation (÷)





Topic : Projection (π)

- It is used to project the column data from a relation based on the attributes specified with projection operation.

e.g., $\pi_{\text{<attribute list>}}(R)$

Name of relation

List of attributes required in o/p

Projection operator need not obey commutative property i.e.

$$\pi_{\text{<list2>}}(\pi_{\text{<list1>}}(R)) \neq \pi_{\text{<list1>}}(\pi_{\text{<list2>}}(R))$$

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

① Retrieve all records from relation R

i.e; all attributes are required in O/p

$\pi_{Sid, Cid, Branch}(R) \equiv (R)$

Both will produce Complete relation R as output

If we do not use Projection opn then all the attributes of relational Schema R will be Present in O/p

② Retrieve Sids of all students from relation R.

$\pi_{Sid}(R) \Rightarrow$ O/p

Sid
S_1
S_2
S_3

③ Retrieve all Cids from relation R

$\pi_{Cid}(R) \Rightarrow$ O/p

Cid
C_1
C_2
~~C_1~~

duplicate tuple will not be present.

④

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

$\pi_{Sid, Cid}(R) \Rightarrow$ o/p

Sid	Cid
S_1	C_1
S_2	C_2
S_3	C_1

tuple wise they are distinct

⑤

$$\pi_{\underbrace{Sid}_{List\ 2}}\left(\pi_{\underbrace{Sid, Cid}_{List\ 1}}(R)\right) \not\equiv \pi_{\underbrace{Sid, Cid}_{list\ 1}}\left(\pi_{\underbrace{Sid}_{list\ 2}}(R)\right)$$

invalid query



Topic : Selection(σ)

- It is used to select the tuples from underlying relation based on the predicate condition specified with selection operation.

$$\sigma_{\text{<selection_condition>}}(R)$$

Name of Relation

Condition to filter the tuples

NOTE:

$$\sigma_{A \wedge B}(R) = \sigma_{B \wedge A}(R)$$

OR

$$\sigma_{B}(\sigma_{A}(R)) = \sigma_{A}(\sigma_{B}(R))$$

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

① $\Pi_{Sid,Cid,Branch}(R) \equiv (R)$

We did not specify any selection,
∴ All the tuples of relation R are selected

② Retrieve records of the students who enrolled for Cource with $Cid = C_1$

i.e; in the o/p all attributes are required

$\sigma_{Cid='C_1'}(R) \Rightarrow$ o/p

Projection is not used, ∴ all attributes will be Present in output.

Sid	Cid	Branch
S_1	C_1	CS
S_3	C_1	CS

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

(3) Retrieve Sids of the students who enrolled for Cource with Cid: C_1

$$\pi_{Sid}\left(\sigma_{Cid='C_1'}(R)\right) \Rightarrow \text{O/p:}$$

Sid
S_1
S_3

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

④ Retrieve records in which Cid is 'C_1' and Branch is 'IT'

$$\sigma_{Cid = 'C_1' \wedge Branch = 'IT'}(R)$$

"AND"

$\Rightarrow$ o/p =

Sid	Cid	Branch

$$\sigma_{Branch = 'IT'}\left(\sigma_{Cid = 'C_1'}(R)\right)$$

Out will be Empty relation.

{ Schema will exist, but no tuple will be present }

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

(5) Retrieve records in which Cid is 'C1' or Branch is 'IT'

$$\sigma_{Cid = 'C_1' \vee Branch = 'IT'}(R)$$

$$\Rightarrow o/p = \left(\sigma_{Branch = 'IT'}(R) \cup \sigma_{Cid = 'C_1'}(R)\right)$$

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS



Topic : NOTE

let 'A' & 'B' are Conditions.

- $\sigma_{A \wedge B}(R) = \sigma_{B \wedge A}(R)$

OR

$\sigma_{B}(\sigma_{A}(R)) = \sigma_{A}(\sigma_{B}(R))$

(OR)

$\sigma_{A}(R) \cap \sigma_{B}(R)$

$\sigma_{B}(R) \cap \sigma_{A}(R)$

- $\sigma_{A \vee B}(R) \equiv \sigma_{B \vee A}(R)$

(or)

$\sigma_{A}(R) \cup \sigma_{B}(R)$

(or)

$\sigma_{B}(R) \cup \sigma_{A}(R)$



Topic : Cross Product (×) / Cross Join / Cartesian Join

- Cross-product is a binary operation. Let R and S are any two relation, then cross product R × S will result in all attributes of R followed by all attribute of S with all possible combinations of tuples from R and S.

  ↳ i.e. each tuple of relation R will join with each tuple of relation S

'x' attributes

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

m tuples

'y' attributes

S

Sid	Sname
S_1	Ram
S_2	Mohan
S_3	Ram

n tuples

$R \times S =$

'x+y' attributes

R.Sid	R.Cid	R.Branch	S.Sid	S.Sname
S_1	C_1	CS	S_1	Ram
S_1	C_1	CS	S_2	Mohan
S_1	C_1	CS	S_3	Ram
S_2	C_2	IT	S_1	Ram
S_2	C_2	IT	S_2	Mohan
S_2	C_2	IT	S_3	Ram
S_3	C_1	CS	S_1	Ram
S_3	C_1	CS	S_2	Mohan
S_3	C_1	CS	S_3	Ram

(m*n) tuples



Topic : Union, Set difference, Intersection

- Union, Set Difference and Intersection are the Set operations.
- To use set theory operators on any two relations, those relations must be union compatible.
- The union compatibility of relations implies that the participating relations must fulfil the following conditions.

1. Same degree, i.e. The two relations must have the same number of attributes.
2. Same domain of each corresponding attributes of relations

the type of values that they can acquire.
(i.e; data type)

R

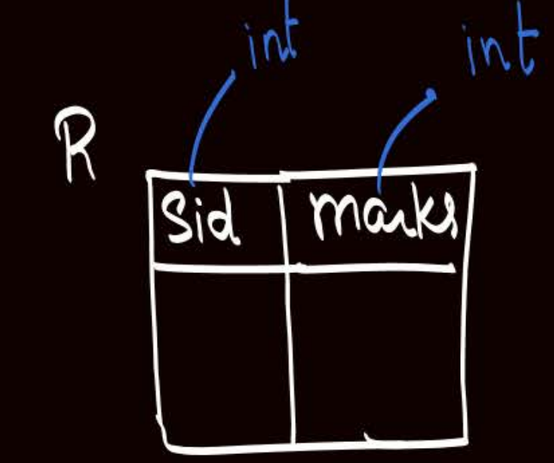
Sid	Cid	Branch

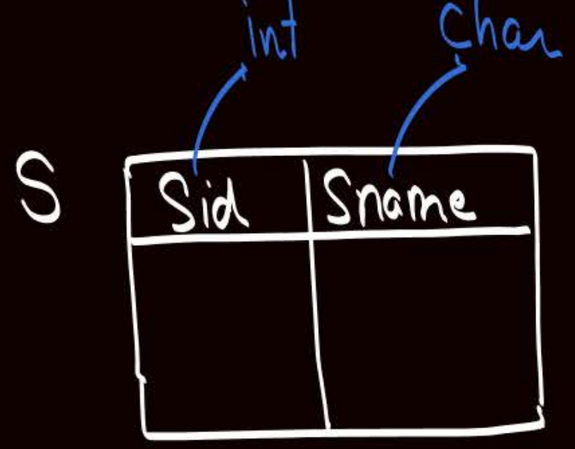
S

Sid	Sname

No. of attributes in R $\neq$ No. of attributes in S

$\therefore$ R & S are not Union Compatible.





- No. of Attributes in R = ✓ No. of attributes in S
- domain of first attribute(Sid) of R = ✓ domain of first attribute(Sid) of S
- domain of 2nd attribute(marks) of R = ✗ domain of 2nd attribute(Sname) of S

  int ≠ Char

∴ R & S are not Union Compatible

R

Sid	Branch

S

Sid	Sname

① No. of attributes in R = ✓ No. of attributes in S

② Domain of 1st attribute of R = Domain of 1st attribute of S

&

Domain of 2nd attribute of R = Domain of 2nd attribute of S

} Same degree ✓ + Domain of each corresponding attribute is same ✓

⇓

∴ R & S are Union Compatible

Note: ① Set operations can be performed on two relations if and only if relations are Union Compatible

② After the set operation the resulting relation will take the names of its attributes from left hand side relation.

i.e, (i) In R ∪ S, names of attributes will be same as names of attributes in Relⁿ R.

and (ii) In S ∪ R, names of attributes will be same as names of attributes in Relⁿ S



Topic : Union, Set difference, Intersection

- Union (A ∪ B)- It contains unique tuples from both the relations.

- Difference (A – B)- It contains all the tuples that are contained in the relation A but are not present in the relation B

- Intersection (A ∩ B)- It contains all the tuples that are contained in both the relations A as well as in relation B.

Employee (E)

Eid	Ename
1	A
3	B
5	A
8	C

Student (S)

Sid	Sname
2	D
3	B
5	A
7	C

E ∪ S =

Eid	Ename
1	A
3	B
5	A
8	C
2	D
7	C

S ∩ E =

S − (S − E)

Sid	Sname
3	B
5	A

S − E =

Sid	Sname
2	D
7	C

E − S =

Eid	Ename
1	A
8	C



2 mins Summary

- **Topic** Query Languages
- **Topic** Relational algebra
- **Topic** Basic relational algebra operations

THANK - YOU